Lecture 11

Homogeneous Linear 2nd-Order Equations

Consider the linear 2nd-order differential equation

$$\underbrace{\overset{\neq 0}{a(t)}y'' + b(t)y' + c(t)y}_{} = \boxed{f(t)}, \quad a(t) \neq 0 \tag{11.1}$$

no y's

If $f(t) = 0$, then the equation

$$\overset{c}{a(t)}y'' + \overset{0}{b(t)}y' + \overset{0}{c(t)}y = \boxed{0} \tag{11.2}$$

homogeneous

is the associated **homogeneous equation** of (11.1).

Note that $y = 0$ is always a solution to (11.2) (called a trivial solution). So we will usually exclude trivial solutions when discussing solving homogeneous equations.

Theorem: If $y_1(t)$ and $y_2(t)$ are solutions to (11.2), then any linear combination

$$y = c_1y_1(t) + c_2y_2(t)$$

is also a solution to (11.2)

assume $\begin{cases} a(t)y_1'' + b(t)y_1' + c(t)y_1 = 0 \\ a(t)y_2'' + b(t)y_2' + c(t)y_2 = 0 \end{cases}$

Proof:

$$y = c_1y_1 + c_2y_2$$
$$y' = c_1y_1' + c_2y_2'$$
$$y'' = c_1y_1'' + c_2y_2''$$

So $a(t)[c_1y_1'' + c_2y_2''] + b(t)[c_1y_1' + c_2y_2'] + c(t)[c_1y_1 + c_2y_2]$

$[a(t)c_1y_1'' + b(t)c_1y_1' + c(t)c_1y_1'] + [a(t)c_2y_2'' + b(t)c_2y_2' + c(t)c_2y_2]$

$c_1\underbrace{[a(t)y_1'' + b(t)y_1' + c(t)y_1]}_{0} + c_2\underbrace{[a(t)y_2'' + b(t)y_2' + c(t)y_2]}_{0} \overset{\checkmark}{=} 0$

assumption

We will be interested in finding collections of solutions to equations. If a solution can be written as a linear combination of another solution then they are essentially the same solution (as in they don't give us additional information about the nature of the equation and solutions to it). We make the following definition to help separate "unique" solutions.

$Cy_1 = y_2$ dependent

Definition: A pair of functions y_1 and y_2 are **linearly independent** on some interval I if neither one is a constant multiple of the other. If they are not linearly independent, we say that are **linearly dependent**. $\Rightarrow C \cdot y_1 - y_2 = 0$

Example: Are the following pairs of functions linearly independent or linearly dependent?

- $y_1(t) = 1,\ y_2(t) = t^{1/2}$ $\quad C \cdot 1 \neq \sqrt{t}$ independent
- $y_1(t) = 3t^{1/2},\ y_2(t) = t^{1/2}$ $\quad 3y_2 = 3 \cdot \sqrt{t} = y_1$ dependent
- $y_1(t) = t,\ y_2(t) = e^t$ $\quad \underset{\text{linear}}{C \cdot t} \neq \underset{\text{exponential}}{e^t}$ independent

If we have two linearly independent solutions y_1 and y_2 to the homogeneous equation

$$a(t)y'' + b(t)y' + c(t)y = 0,$$

then

$$\boxed{y_h = C_1 y_1 + C_2 y_2}$$

homogeneous

is its **general solution** where C_1 and C_2 are constants.

we will show that a 2nd-order linear equation has exactly 2 linearly independent solutions

In the previous lecture we began studying 2nd-order equations of the form

$$ay'' + by' + cy = f(t), \quad a \neq 0$$

constants

which we call a **constant coefficient** equation. What form do the solutions to the constant coefficient equation take?

Recall, the constant coefficient 1st-order linear equation is

$$\underbrace{ay' + by = 0}_{\text{separable}}, \quad a \neq 0$$

$$y' = -\frac{by}{a} \Rightarrow \frac{dy}{dt} = -\frac{b}{a}y \Rightarrow \frac{1}{y}\,dy = -\frac{b}{a}\,dt$$

$$\Rightarrow \int \frac{1}{y}\,dy = \int -\frac{b}{a}\,dt \Rightarrow \ln|y| = -\frac{b}{a}t + C$$

$$\Rightarrow y = e^{-\frac{b}{a}t + C} = \underline{\underline{C}}\,\underbrace{e^{-\frac{b}{a}t}}_{\text{form}}$$

Therefore, we may want to try the function $\underline{\underline{y = e^{rt}}}$ as a solution to

$$ay'' + by' + cy = 0$$

$$y = e^{rt}, \quad y' = re^{rt}, \quad y'' = r^2e^{rt}$$

substitute

$$a(r^2e^{rt}) + b(re^{rt}) + ce^{rt} \overset{\text{want}}{=} 0$$

$$\Rightarrow \underbrace{e^{rt}}_{\neq 0}\underbrace{[ar^2 + br + c]}_{\text{need } = 0} = 0$$

$\boxed{e^x > 0}$

If $y = e^{rt}$ is a solution, then $\boxed{\underbrace{ar^2 + br + c}_{r\text{-variable}} = 0}$

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \qquad r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

($b^2 - 4ac$ determined the solutions)

The equation

$$ar^2 + br + c = 0$$

is called the **characteristic** (or **auxiliary**) **equation** associated to $ay'' + by' + cy = 0$.

mass-spring b^2-4mk

The form of the solution is determined by $b^2 - 4ac$.

there are a few cases depending upon the value of b^2-4ac

Case 1: $b^2 - 4ac > 0$, two distinct roots $r_1 \neq r_2$.

$r_1 = \frac{-b+\sqrt{b^2-4ac}}{2a}$, $r_2 = \frac{-b-\sqrt{b^2-4ac}}{2a}$ real numbers

$r_1 \neq r_2$

Two linearly independent solutions are $e^{r_1 t}, e^{r_2 t}$

The general solution is $y_h = c_1 e^{r_1 t} + c_2 e^{r_2 t}$

Example: Consider the equation

$$y'' + y' - 6y = 0$$

$a=1, b=1, c=-6$

characteristic eqn: $ar^2+br+c=0 \Rightarrow r^2+r-6=0$

$\Rightarrow (r-2)(r+3)=0$

$\Rightarrow r = 2, -3$ distinct

So $\boxed{y_h = c_1 e^{2t} + c_2 e^{-3t}}$

Case 2: $b^2 - 4ac = 0$, two repeated real roots

$r = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{-b}{2a}$

$ar^2+br+c = (r+\frac{b}{2a})^2 = 0$

prove this is valid.

We can use $y_1 = e^{rt}$, $y_2 = te^{rt}$ linearly independent

multiply by t.

$y_2 = te^{rt}$, $y_2' = e^{rt} + rte^{rt}$

$y_2'' = re^{rt} + re^{rt} + r^2te^{rt} = (2r+r^2t)e^{rt}$

$a[2r+r^2t]e^{rt} + b[e^{rt}+rte^{rt}] + cte^{rt}$

$= e^{rt}[\underbrace{2ar+b}_{=0}] + te^{rt}[\underbrace{ar^2+br+c}_{=0}] = 0$

so the general solution is $y_h = c_1 e^{rt} + c_2 t e^{rt}$

Example: Consider the equation

$$y'' + 2y' + y = 0$$

$a=1,\ b=2,\ c=1 \qquad b^2-4ac = 4-4(1)(1)=0$ ✓

$r^2+2r+1=0$

$\Rightarrow (r+1)^2=0 \Rightarrow r=-1,-1$

So $\boxed{y_h = c_1 e^{-t} + c_2 t\cdot e^{-t}}$

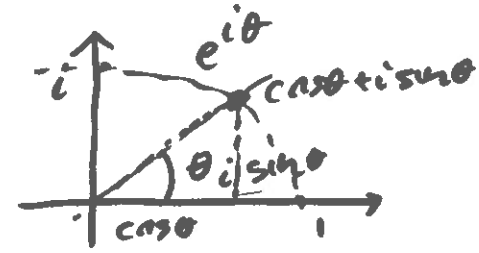
Case 3: $b^2 - 4ac < 0$, complex conjugate pair of roots

$r = \dfrac{-b \pm \sqrt{b^2-4ac}}{2a}$ ← negative imaginary.

$r = \alpha \pm i\beta,\ \alpha = -\dfrac{b}{2a},\ \beta = \dfrac{\sqrt{4ac-b^2}}{2a}$

$r = -\dfrac{b}{2a} \pm i\dfrac{\sqrt{4ac-b^2}}{2a},\ y_1 = e^{(\alpha+i\beta)t} = e^{\alpha t} e^{i\beta t}$

We see that the (complex) solution is of the form $e^{\alpha t}e^{i\beta t}$. We need find solutions that are real-valued functions. To do this we will convert $e^{i\beta t}$ to another form.



Theorem (Euler's Formula): $\boxed{e^{i\theta} = \cos(\theta) + i\sin(\theta)}$

Proof: Assuming the Maclaurin series for $e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$. Suppose that $z = i\theta$ along with

$$i^1 = i, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1, \quad i^5 = i$$

we see that

$$
\begin{aligned}
e^{i\theta} = \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!} &= 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \cdots \frac{(i\theta)^n}{n!} + \cdots \\
&= 1 + i\theta - \frac{\theta^2}{2!} - i\frac{\theta^3}{3!} + \frac{\theta^4}{4!} + i\frac{\theta^5}{5!} + \cdots \\
&= \underbrace{\left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} + \cdots\right)}_{\text{even}} + i\overbrace{\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \cdots\right)}^{\text{odd}} \\
&= \sum_{n=0}^{\infty} \frac{(-1)^n \theta^{2n}}{\underset{\text{evens}}{(2n)!}} + i \sum_{n=0}^{\infty} \frac{(-1)^n \theta^{2n+1}}{\underset{\text{odds}}{(2n+1)!}} \\
&= \cos(\theta) + i\sin(\theta)
\end{aligned}
$$

Note that $e^{-i\theta} = \cos(\theta) - i\sin(\theta)$ which is the complex conjugate of $\cos(\theta) + i\sin(\theta)$.

Therefore we can say that

$$e^{(\alpha+i\beta)t} = e^{\alpha t}e^{i\beta t} = e^{\alpha t}(\cos(\beta t) + i\sin(\beta t))$$

To find the real-valued solutions of $ay'' + by' + cy = 0$ we note that if $u(t) + iv(t)$ is a solution to the equation then both $u(t)$ and $v(t)$ are also solutions.

$$z(t) = \underbrace{e^{\alpha t}\cos(\beta t)}_{u} + i\,\underbrace{e^{\alpha t}\sin(\beta t)}_{v}$$

$z = u(t) + iv(t)$

$z' = u'(t) + iv'(t)$

$z'' = u''(t) + iv''(t)$

$a(u'' + iv'') + b(u' + iv') + c(u + iv) = 0$ solution

$$\underbrace{[au'' + bu' + cu]}_{=0} + i\underbrace{[av'' + bv' + cv]}_{=0} = 0.$$

$a + ib = 0$

$\Rightarrow a, b = 0$

So u, v are solutions!

As $\cos(\beta t)$ and $\sin(\beta t)$ are linearly independent we have that

$$\boxed{y_h = C_1 e^{\alpha t}\cos(\beta t) + C_2 e^{\alpha t}\sin(\beta t)}$$

is a general solution to $ay'' + by' + cy = 0$ if $\alpha + i\beta$ is a root of the corresponding characteristic equation.

Example: Consider the equation

$$y'' + 9y = 0$$

$a=1,\quad b=0,\ c=9 \qquad b^2-4ac = -36 < 0$

characteristic eqn: $r^2+9=0 \Rightarrow r = \pm 3i$

$\longrightarrow r^2 = -9 \rightarrow$

$r = \alpha + i\beta = 0 + i\cdot 3$

$\alpha = 0,\ \beta = 3$

So $y_h = c_1 e^{0\cdot t}\cos(3t) + c_2 e^{0\cdot t}\sin(3t)$

$= c_1\cos(3t) + c_2\sin(3t)$

Example: Consider the equation

$$y'' + 2y' + 4y = 0$$

$a=1,\ b=2,\ c=4 \qquad b^2-4ac = 4-4(1)(4)$

$= 4-16 = -12 < 0$

char. eqn: $r^2+2r+4=0$

$r = \dfrac{-2 \pm \sqrt{-12}}{2(1)} = \dfrac{-2 \pm i\cdot 2\sqrt{3}}{2} = -1 \pm i\sqrt{3}$

$\alpha = -1,\ \beta = \sqrt{3}$

So $y_h = c_1 e^{-t}\cos(\sqrt{3}t) + c_2 e^{-t}\sin(\sqrt{3}t)$

The techniques we have developed for 2nd-order equations applies to higher order equations by constructing an associated auxiliary equation.

Example: Consider the equation

$$y''' + y'' - 6y' + 4y = 0$$

use $\left.\begin{array}{l} y = e^{rt} \\ y' = re^{rt} \\ y'' = r^2e^{rt} \\ y''' = r^3e^{rt} \end{array}\right\} \longrightarrow$ char. eqn: $1 \cdot r^3 + r^2 - 6r + 4 = 0$

<u>rational roots theorem</u>

$ax^n + \cdots + b = 0$

a: highest order coefficient; b: constant

a rational root must be of the form

$$\pm \frac{\text{factors of } b}{\text{factors of } a}$$

$\underbrace{1}_{a} \cdot r^3 + r^2 - 6r + \underbrace{4}_{b} = 0$

factors of $4 = 1, 2, 4$

factors of $1 = 1$

rational roots are $\pm 1, 2, 4$

$1^3 + 1^2 - 6(1) + 4 = 6 - 6 = 0$ ✓ $r = 1$ root

$$\begin{array}{r} r^2 + 2r - 4 \\ r - 1 \overline{\big) r^3 + r^2 - 6r + 4} \\ -(r^3 - r^2) \\ 2r^2 \\ -(2r^2 - 2r) \\ -4r + 4 \\ -(-4r + 4) \\ 0 \end{array}$$

so $r^3 + r^2 - 6r + 4 = (r-1)(r^2 + 2r - 4)$

so $r = 1$, $r = \dfrac{-2 \pm \sqrt{4 - 4(1)(-4)}}{2}$

or $r = 1$, $r = \dfrac{-2 \pm \sqrt{20}}{2}$

$= \dfrac{-2 \pm 2\sqrt{5}}{2} = -1 \pm \sqrt{5}$

so $\boxed{y_h = c_1 e^{\overset{r=1}{t}} + c_2 e^{\underset{r=-1+\sqrt{5}}{(-1+\sqrt{5})t}} + c_3 e^{\underset{r=-1-\sqrt{5}}{(-1-\sqrt{5})t}}}$